

# Question

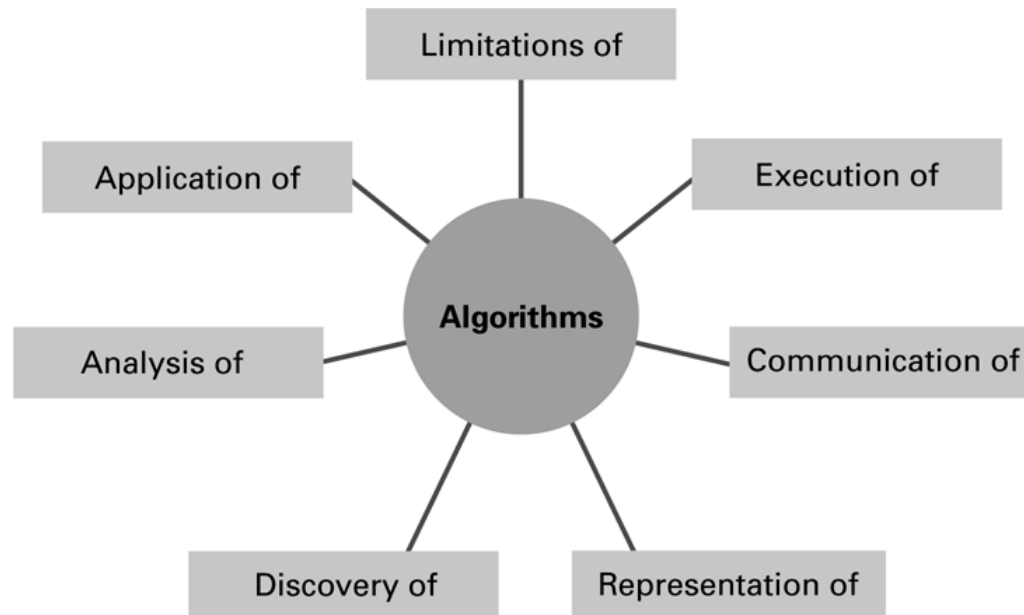
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Explain the importance of algorithms in the field of computer science.

# Figure 0.5 The central role of algorithms in computer science

## The science of algorithms

Computer science has established itself as the science of algorithms. The scope of this science is broad, drawing from such diverse subjects as mathematics, engineering, psychology, biology, business administration, and linguistics. The study of algorithms includes limitations, execution, communication, representation, discovery, analysis, application of algorithms.



# Question

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What's the difference between algorithm and programming?

# Terminology

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**Algorithm:** A set of steps that defines how a task is performed

**Program:** A representation of an algorithm

**Programming:** The process of developing a program

**Software:** Programs and algorithms

**Hardware:** Equipment

# Question

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What's the difference between computer engineering and Computer science?

# Question

- What's the difference
  - Computer engineering
    - Learning to use the application software
  - Computer science
    - Deals with the development of tomorrow's application software

# Question

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What's the difference between abstract and abstract tool?

# Abstraction

- **Abstraction:** The distinction between the external properties of an entity and the details of the entity's internal composition
- **Abstract tool:** A “component” that can be used without concern for the component's internal properties



- What are the seven “Big **I**deas” that unite computer science?

- Seven “Big Ideas” that unite computer science:
  - Algorithms, Abstraction, Creativity, Data, Programing, Internet and Impact